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MS Artificial Intelligence (5015872)

S014_L_vs_R_hand

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1. Introduction

Brain-computer interfaces (BCIs) are a promising technology for connecting the brain and devices. Motor imagery (MI) is a form of conscious access to action representations that does not involve overt motor output (Louis et al., 2011). Conscious action without muscle movement makes MI a widely used control signal for electroencephalography (EEG), with numerous applications in rehabilitation, learning, and virtual reality.

When imagining a movement, synchronized neural firing within the mu (8-12 Hz) and beta (13-30 Hz) frequency bands decreases. This is known as event-related desynchronization (ERD). ERD causes a distinct drop in spectral power relative to the pre-stimulus baseline and persists throughout the mental imagery task (Pfurtscheller & Lopes da Silva, 1999). At the end of the imagery task, the sensorimotor cortex shows an increase in power above baseline level. This surge is known as event-related synchronization (ERS) or "beta rebound" (Pfurtscheller & Neuper, 2001).

Spatially, the central scalp region over the primary motor cortex measures the activity related to motor movement and imagery via ERD and ERS. The electrodes C3, Cz, and C4 in the 10-20 EEG system are used for this purpose. Right-hand movement/imagery causes ERD primarily at C3 and left-hand movement/imagery at C4.

These spatial and spectral differences form a physiological basis for differentiating left- and right-hand motor imagery from scalp EEG using spatial filtering methods. Algorithms such as Common Spatial Patterns (CSP) learn linear combinations that maximize variance differences between classes in a designated frequency band, and classifiers such as Linear Discriminant Analysis (LDA) can classify the imagery action (Blankertz et al., 2008; Ramoser et al., 2000).

Building on the above-mentioned framework, the report poses the following research question: can a single-trial left versus right hand motor imagery be differentiated from EEG using Common Spatial Patterns (CSP) feature extraction and a linear classifier such as Linear Discriminant Analysis (LDA)? The hypothesis is that left- and right-hand imagery will produce spatially and spectrally lateralized mu/beta ERD over C3 and C4, and classification accuracy will be above the 50% chance level.

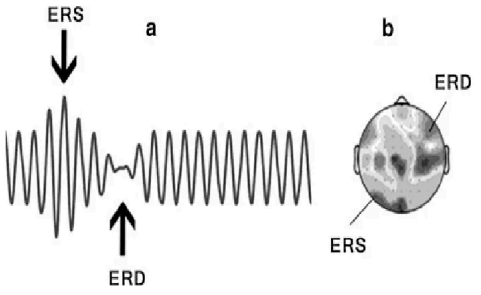


Figure 1: a) ERD/ERS components in an observed brain signal; b) Topographical map of ERD/ERS components in a hand movement motor imagery (Niedermeyer & Lopes da Silva, 2005)

2. Methods

2.1 Experimental Paradigm

The experiment is a subset (S014) of the PhysioNet EEG Motor Movement/Imagery Database (Goldberger et al., 2000; Schalk et al., 2004). For the subset data, 64-channel EEG was recorded using the BCI2000 system (http://www.bci2000.org) while the participants performed the following motor/imagery tasks: a) a target appears on either the left or the right side of the screen. The subject opens and closes the corresponding fist until the target disappears. Then the subject relaxes. b) A target appears on either the left or the right side of the screen. The subject imagines opening and closing the corresponding fist until the target disappears. Then the subject relaxes.

Each subject performed 14 experimental runs: two one-minute baseline runs (one with eyes open, one with eyes closed), and three two-minute runs of each task. Each event code includes an event marker: T0 for rest, T1 for imagining a left fist, and T2 for imagining a right fist (~4 s each).

2.2 Data Acquisition

EEGs were recorded from 64 electrodes as per the international 10-10 system (excluding electrodes Nz, F9, F10, FT9, FT10, A1, A2, TP9, TP10, P9, and P10), as shown in the figure. The numbers below each electrode name indicate the order in which they appear in the records; note that signals in the records are numbered from 0 to 63, while the numbers in the figure range from 1 to 64. The EEG cap is from Electro-Cap International, Inc. The reference used is the left or right earlobe. The power line frequency is 60 Hz, and the sampling frequency is 160 Hz.

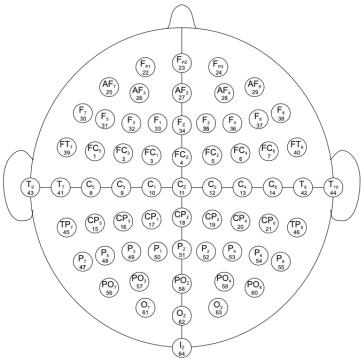


Figure 2: 64-channel EEG placements with their order number

2.3 Dataset

The data has 45 trials from a 26-year-old, right-handed male. 22 trials have event marker T1 for left-handed imagery, and 23 trials have T2 for the right-handed imagery. The recording consists of three concatenated runs, is continuous, and is 396 s (~6.2 minutes).

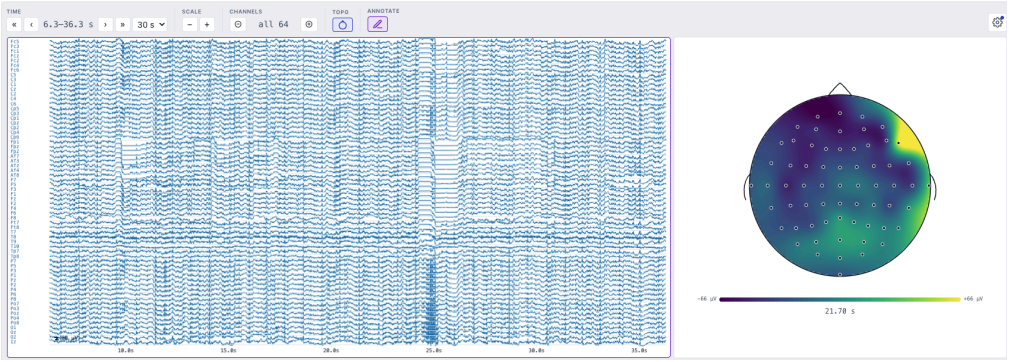


Figure 3: An overview of the EEG data with a topographical map

2.4 EEG Preprocessing

The electrodes were used to add their locations to assist in later steps such as ICA and topographic maps. A 1 Hz high-pass filter removed slow drifts, while a 40 Hz low-pass filter removed high frequencies, retaining mu (8-12 Hz) and beta (13-30 Hz) rhythms. clean_artifacts function from the EEGLAB plugin removed bad channels and performed burst correction. The values were based on previous research (Mullen et al., 2015). Interpolation helped in recovering any ASR-removed channels.

ICA decomposition checked signal sources relating to the brain, eyes, and muscles. Muscle and eye components are removed with a probability greater than 80%. Automatic, reproducible artifact detection (Pion-Tonachini et al., 2019). As a result, 7 of 63 components, such as IC1, 4, 10, 11, 18, 23, 38, are removed. The highest eye-muscle-probability components were removed, whereas the highest brain-probability components were kept.

IC1 (eye, p=0.999)**:** There is a sharp rise and then a fall around t=13, which is a classic eye blink. It is also below 4 Hz with no alpha/beta structure. The scalp map also shows activity near the eye region. It can be classified as an eye artifact and hence removed.

IC10 (muscle, p=0.976)**:** It has continuous high amplitude and high frequency with no slow structure to assume ERD. The frequency stays elevated at 15-40 Hz. The scalp map shows a focal hotspot over the left ear. A cortical source is usually smooth. It can be classified as a jaw/ muscle artifact and removed.

IC3 & IC9 (brain, p=0.999): Both are irregular and show moderate fluctuations. Both have clear peaks in the 10-12 Hz range, which can be the alpha-mu rhythm. The scalp map also shows a smooth distribution in both cases. This implies a brain rhythm; therefore, IC3 and IC9 were kept.

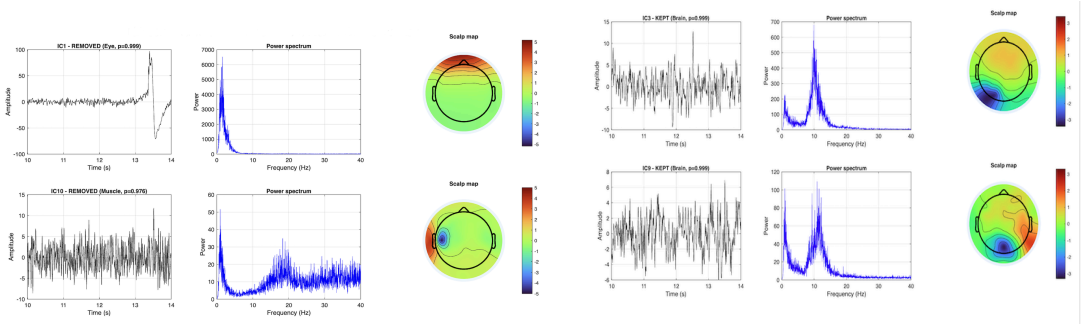


Figure 4: Time course, power spectrum, and scalp map for each ICA component: IC1 (removed, eye) and IC10 (removed, muscle) vs. IC3 and IC9 (both kept, brain)

The epoch window is -1 s to +4 s around each cue, with baseline correction on -1000 to 0 ms.

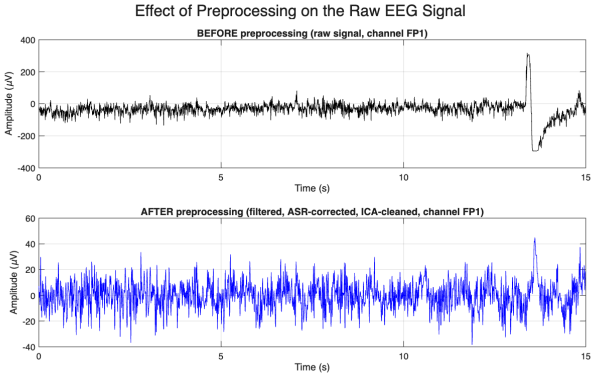


Figure 5: Effect of preprocessing on the raw EEG signal

2.5 BCI Approach

BCILAB Common Spatial Patterns (CSP) algorithm was applied for feature extraction. CSP constructs spatial filters that maximize power for one condition while minimizing it for the other (Ramoser et al., 2000). For motor imagery tasks, this filtered signal variance reflects the neural power within the sensorimotor rhythms. By calculating spatial projections tuned to 8-30 Hz, CSP can isolate the contralateral mu and beta event-related desynchronization (ERD) over electrode sites C3 and C4, reducing the spatial blur caused by volume conduction across the scalp.

Linear discriminant analysis (LDA) was used for classification because CSP-extracted features are low dimensional and normally distributed. LDA provides a linear decision boundary while avoiding overfitting. The CSD+LDA combination is a standard baseline for two-class motor imagery classification in BCI (Blankertz et al., 2008).

2.6 Classifier training

Cue markers T1 and T2 were used for imagined left-hand movement and imagined right-hand movement, respectively. There were 22 trials for T1 and 23 trials for T2, making it a total of 45 trials and an even class split. This gives 50% as a fair chance baseline. The rest marker (T0) was not used for classification.

After removing the artifacts, the continuous data was provided to bci_train in a single call. This training procedure includes band-pass filtering, extracting epochs, computing CSP spatial filters from training trials, extracting log-variance features, and fitting an LDA on the acquired features.

BCILAB 5-fold chronological cross-validation was used to evaluate the model. The continuous data were divided into five time-ordered blocks. Five trials were excluded around each test/train boundary to reduce signal leakage and correlation for adjacent trials. The misclassification rate (mcr) and accuracy were averaged across five held-out folds to minimize overfitting.

3. Results

3.1 EEG Patterns of Motor Imagery (Left vs. Right Hand)

The event-related potential (ERP) waveforms at C3 and C4 show no consistent voltage deflections time-locked to cue onset (t=0). Across the 4 s epoch window, signals from left- and right-hand imagery overlap, forming high-variance spikes as shown in the figure.

It is expected to have a flat ERP wave in this case because motor imagery does not evoke phase-locked stimulus response. Instead, it manifests as a change in power in ongoing sensorimotor oscillations (Pfurtscheller & Lopes da Silva, 1999).

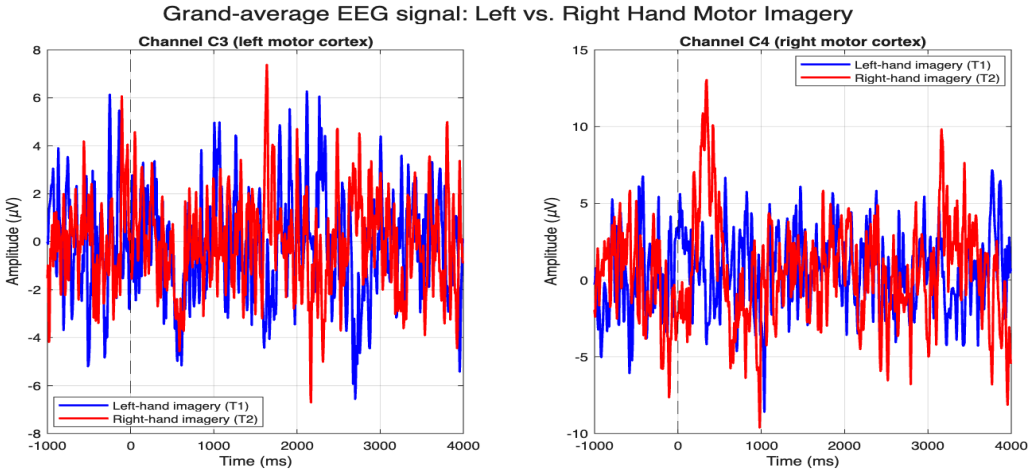


Figure 6: Averaged signal: left vs. right hand motor imagery

The time-frequency domain, i.e., Event-Related Spectral Perturbation (ERSP), shows the neural dynamics more distinctly in this case. At C3, both conditions show a power decrease (blue, -2 to -3 dB) concentrated in the 8- 15 Hz range. It begins shortly after 0 ms and extends roughly to 500-3000ms. This confirms event-related desynchronization (ERD) by overlap of the mu band and the start of the beta band during the imagery period. At C4, a similar and more fragmented ERD was observed in the 8-20 HZ passband.

The inter-trial phase coherence (ITC) was uniformly low and unstructured; hence, the signal is not another evoked potential.

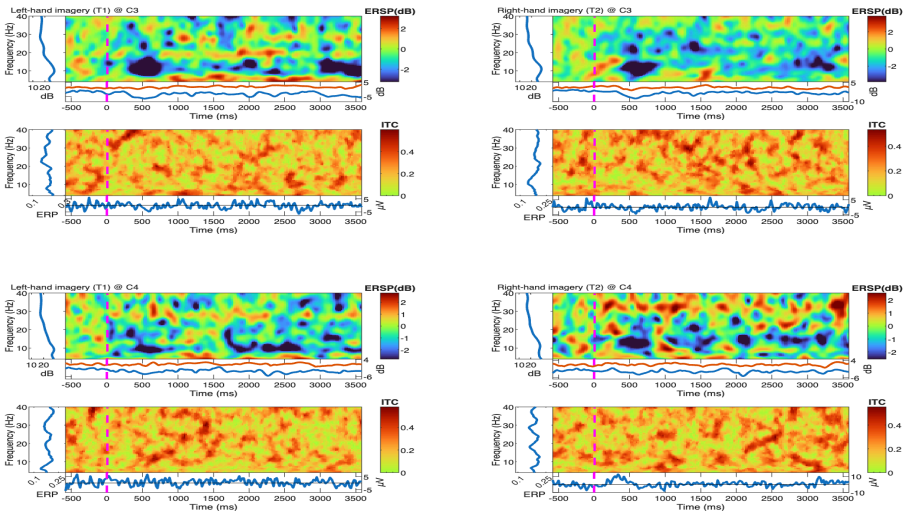


Figure 7: ERPS of C3 and C4 for the left and right hand motor imagery

The scalp topography doesn't have a clear indication because the mu-band (8-12 Hz) is overshadowed by a large posterior region in both conditions.

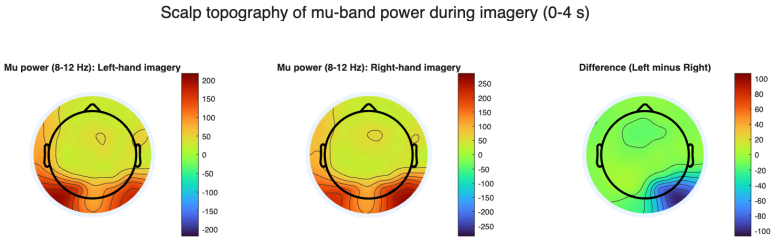


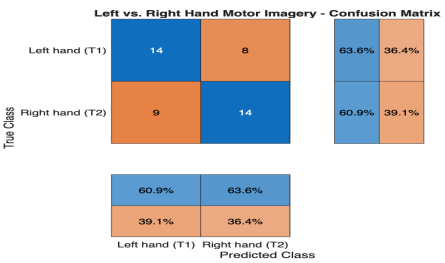
Figure 8: Scalp topography for left and right hand imagery (mu power)

The time and frequency domains match the literature, but the spatial lateralization is not visible. This implies a partial match for confirming the target signal. This also aligns with the accuracy range achieved, i.e., 55- 65% for the CSP+LDA classifier.

3.2 BCI Model Validation

The overall accuracy for CSP with LDA was around 62% (28/45 trials were correctly classified) across the 5-fold chronological cross-validation procedure, as shown in the table below. The symmetry between class metrics shows unbiasedness. Since both classes have nearly equal numbers of trials, one baseline is 50%. 62.2% is 12.2 points above the 50% baseline. It sits at 1.5-1.6 standard errors above chance, making it a noisy discriminative signal.

Condition	Recall	Precision
Left hand (T1)	63%	61%
Right hand (T2)	61%	63%
Overall accuracy	62%	



An empirical chance level with a mean of 49.1% was also calculated with 20 label-shuffled permutations. Our accuracy was higher in 2 of 20 permutations. This implies that the model accuracy doesn't reach conventional statistical significance. It might improve with more permutations.

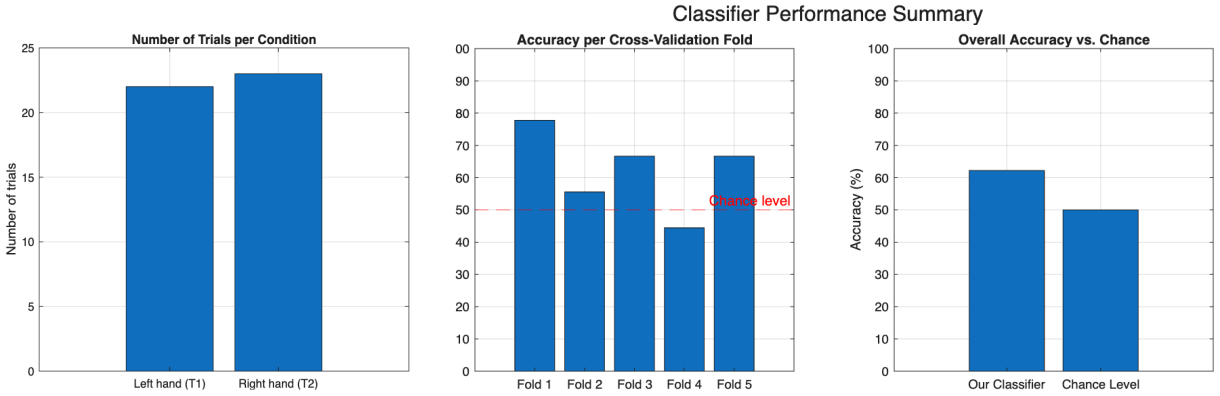


Figure 9: a) class distribution, b) accuracy per class validation fold, c) overall accuracy and chance level

Upon checking the CSP spatial patterns, the following six were fed to the LDA classifier:

Pattern	Top channels	Region
1	PO8, P8, PO4, O2, P6	Posterior/occipital
2	PO4, O2, PO8, Oz, POz	Posterior/occipital
3	POz, Pz, P1, P3, PO3	Posterior/parietal
4	PO3, P5, P7, CP5, TP7	Posterior/parietal
5	PO7, P7, P5, O1, PO3	Posterior/occipital
6	C6, F3, C2, F1, AFz	Central-right + frontal

It was disappointing to see that the classifier did not base its decision on the expected central sensorimotor channel. Five of six CSP patterns are dominated by parieto-occipital electrodes. Therefore, the 62% accuracy should be interpreted cautiously. The artefacts were removed, but the model was more based on posterior/parieto-occipital channels.

4. Discussion

The research question states whether a single-trial EEG could be classified as left- vs. right-hand motor imagery using CSP and LDA. Quantitatively, the accuracy (62%) remains above chance level for both normal and empirical cases. Checking the channels influencing the classifier makes this result statistically insignificant. The first hypothesis, accuracy being above 50%, is satisfied. However, the second hypothesis, that left- and right-hand imagery

will produce spatially and spectrally lateralized mu/beta ERD over C3 and C4, is partially true.

The ERPS shows a power decrease in the 8-20 Hz range within ~0.5 s of cue onset. But scalp topography was dominated by the posterior alpha rather than a central collateral dipole, which was also reflected in feature bias. Therefore, the classifier can be assumed weakly effective.

One limitation is that CSP is spatially agnostic. Because posterior alpha has a larger amplitude, CSP gravitated toward it more. The sample included one person and one session, making it harder to run calculations like permutations. These limitations can be minimized by using filter bank CSP or a subject-specific frequency band rather than a fixed one. More trials can be included. A permutation test with more shuffles will provide a more reliable significance estimate.

Real applications can provide a control channel for people who have limited muscle output, for example, locked-in syndrome, spinal cord injury, or late-stage ALS (Wolpaw et al., 2002). It can also support smart devices like motor imagery-controlled wheelchairs, robotics, and more. With the recent boom in AI, it can also be used in neuroadaptive AI technology that relies on motor imagery and passive BCIs.

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